

Write all program in C using Structures**Question 1:**

Build an Student record system that stores.

- Store student name, roll number, marks in 3 subjects
- Calculate percentage and grade based on marks
- Implement a grading system: A+(90-100), A(80-89), B+(70-79), B(60-69), C(50-59), F(<50)
- Add functionality to find students by grade
- Calculate class average and display students above/below average

Additional features:

- Input validation for marks (0-100)
- Search students by roll number or name
- Display student rank based on percentage

Student Name: _____ Wajiha Nazir _____ Roll No: _____ 25K-3021 _____

Section: BSE-1B

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

#define MAX_STUDENTS 100

typedef struct {
    char name[50];
    int roll;
    int marks[3];
    float percentage;
    char grade;
} Student;

Student students[MAX_STUDENTS];
int student_count = 0;
char calculateGrade(float percentage);
void addStudent();
void displayStudent(Student s);
void displayAllStudents();
void searchByRollOrName();
void findByGrade();
void classAverage();
void rankStudents();
int comparePercentage(const void *a, const void *b);

int main() {
    int choice;
    do {
        printf("\n--- Student Record System ---\n");
        printf("1. Add Student\n");
        printf("2. Display All Students\n");
        printf("3. Search Student by Roll or Name\n");
        printf("4. Find Students by Grade\n");
        printf("5. Class Average & Above/Below Average\n");
        printf("6. Display Students by Rank\n");
        printf("0. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch(choice) {
            case 1: addStudent(); break;
            case 2: displayAllStudents(); break;
            case 3: searchByRollOrName(); break;
            case 4: findByGrade(); break;
```

Student Name: _____ Wajiha Nazir _____ Roll No: _____ 25K-3021 _____

Section: BSE-1B

```
        case 5: classAverage(); break;
        case 6: rankStudents(); break;
        case 0: printf("Exiting...\n"); break;
        default: printf("Invalid choice! Try again.\n");
    }
} while(choice != 0);

return 0;
}

char calculateGrade(float percentage) {
    if(percentage >= 90) return 'A';
    else if(percentage >= 80) return 'B';
    else if(percentage >= 70) return 'C';
    else if(percentage >= 60) return 'D';
    else if(percentage >= 50) return 'E';
    else return 'F';
}

void addStudent() {
    int i;
    if(student_count >= MAX_STUDENTS) {
        printf("Cannot add more students!\n");
        return;
    }
    Student s;
    printf("Enter student name: ");
    getchar();
    fgets(s.name, 50, stdin);
    s.name[strcspn(s.name, "\n")] = 0;
    printf("Enter roll number: ");
    scanf("%d", &s.roll);
    for(i = 0; i < 3; i++) {
        do {
            printf("Enter marks for subject %d (0-100): ", i+1);
            scanf("%d", &s.marks[i]);
            if(s.marks[i] < 0 || s.marks[i] > 100)
                printf("Invalid marks! Please enter between 0 and 100.\n");
        } while(s.marks[i] < 0 || s.marks[i] > 100);
    }
    s.percentage = (s.marks[0] + s.marks[1] + s.marks[2]) / 3.0;
    s.grade = calculateGrade(s.percentage);
}
```

Student Name: _____ Wajiha Nazir _____ Roll No: _____ 25K-3021 _____

Section: BSE-1B

```
    students[student_count++] = s;
    printf("Student added successfully!\n");
}

void displayStudent(Student s) {
    printf("Name: %s, Roll: %d, Marks: [%d, %d, %d], Percentage: %.2f, Grade: %c\n",
           s.name, s.roll, s.marks[0], s.marks[1], s.marks[2], s.percentage, s.grade);
}

void displayAllStudents() {
    int i;
    if(student_count == 0) {
        printf("No students to display.\n");
        return;
    }
    for(i = 0; i < student_count; i++)
        displayStudent(students[i]);
}

void searchByRollOrName() {
    if(student_count == 0) {
        printf("No students to search.\n");
        return;
    }
    int choice;
    printf("Search by 1. Roll 2. Name: ");
    scanf("%d", &choice);
    if(choice == 1) {
        int i;
        int roll;
        printf("Enter roll number: ");
        scanf("%d", &roll);
        for(i = 0; i < student_count; i++)
            if(students[i].roll == roll)
                displayStudent(students[i]);
    } else if(choice == 2) {
        int i;
        char name[50];
        printf("Enter name: ");
        getchar();
        fgets(name, 50, stdin);
        name[strcspn(name, "\n")] = 0;
    }
}
```

Student Name: Wajiha Nazir Roll No: 25K-3021

Section: BSE-1B

```
        for(i = 0; i < student_count; i++)
            if(strcmp(students[i].name, name) == 0)
                displayStudent(students[i]);
        } else {
            printf("Invalid choice!\n");
        }
    }

void findByGrade() {
    int i;
    if(student_count == 0) {
        printf("No students to search.\n");
        return;
    }
    char grade;
    printf("Enter grade (A-F): ");
    getchar();
    scanf("%c", &grade);
    for(i = 0; i < student_count; i++)
        if(students[i].grade == grade)
            displayStudent(students[i]);
}

void classAverage() {
    int i;
    if(student_count == 0) {
        printf("No students to calculate average.\n");
        return;
    }
    float total = 0;
    for(i = 0; i < student_count; i++)
        total += students[i].percentage;
    float avg = total / student_count;
    printf("Class Average: %.2f\n", avg);
    printf("Students above average:\n");
    for(i = 0; i < student_count; i++)
        if(students[i].percentage > avg)
            displayStudent(students[i]);
    printf("Students below average:\n");
    for(i = 0; i < student_count; i++)
        if(students[i].percentage < avg)
            displayStudent(students[i]);
}
```

Student Name: _____ Wajiha Nazir _____ Roll No: _____ 25K-3021 _____

Section: BSE-1B

```
printf("Students below average:\n");
for(i = 0; i < student_count; i++)
    if(students[i].percentage < avg)
        displayStudent(students[i]);
}

int comparePercentage(const void *a, const void *b) {
    Student *s1 = (Student *)a;
    Student *s2 = (Student *)b;
    if(s2->percentage > s1->percentage) return 1;
    else if(s2->percentage < s1->percentage) return -1;
    else return 0;
}

void rankStudents() {
    int i;
    if(student_count == 0) {
        printf("No students to rank.\n");
        return;
    }

    qsort(students, student_count, sizeof(Student), comparePercentage);

    printf("Ranked Students:\n");
    for(i = 0; i < student_count; i++) {
        printf("Rank %d: ", i+1);
        displayStudent(students[i]);
    }
}
```

Student Name: _____ Wajiha Nazir _____ Roll No: _____ 25K-3021 _____

Section: BSE-1B

```
--- Student Record System ---
1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit
Enter your choice: 1
Enter student name: Wajiha Nazir
Enter roll number: 3021
Enter marks for subject 1 (0-100): 99
Enter marks for subject 2 (0-100): 55
Enter marks for subject 3 (0-100): 68
Student added successfully!

--- Student Record System ---
1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit
Enter your choice: 1
Enter student name: Rumaisa Nazir
Enter roll number: 3082
Enter marks for subject 1 (0-100): 76
Enter marks for subject 2 (0-100): 85
Enter marks for subject 3 (0-100): 64
Student added successfully!

--- Student Record System ---
1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit
Enter your choice: 1
Enter student name: Tayyaba Nazir
Enter roll number: 3096
Enter marks for subject 1 (0-100): 55
Enter marks for subject 2 (0-100): 23
Enter marks for subject 3 (0-100): 41
Student added successfully!

--- Student Record System ---
1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit
Enter your choice: 1
Enter student name: Zunnera Nazir
Enter roll number: 3013
Enter marks for subject 1 (0-100): 23
Enter marks for subject 2 (0-100): 32
Enter marks for subject 3 (0-100): 41
```

Student Name: Wajiha Nazir Roll No: 25K-3021

Section: BSE-1B

Student added successfully!

--- Student Record System ---

1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit

Enter your choice: 2

Name: Wajiha Nazir, Roll: 3021, Marks: [99, 55, 68], Percentage: 74.00, Grade: C
Name: Rumaisa Nazir, Roll: 3082, Marks: [76, 85, 64], Percentage: 75.00, Grade: C
Name: Tayyaba Nazir, Roll: 3096, Marks: [55, 23, 41], Percentage: 39.67, Grade: F
Name: Zunnera Nazir, Roll: 3013, Marks: [23, 32, 41], Percentage: 32.00, Grade: F

--- Student Record System ---

1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit

Enter your choice: 3

Search by 1. Roll 2. Name: 1

Enter roll number: 3096

Name: Tayyaba Nazir, Roll: 3096, Marks: [55, 23, 41], Percentage: 39.67, Grade: F

--- Student Record System ---

1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit

Enter your choice: 3

Search by 1. Roll 2. Name: 2

Enter name: Rumaisa Nazir

Name: Rumaisa Nazir, Roll: 3082, Marks: [76, 85, 64], Percentage: 75.00, Grade: C

--- Student Record System ---

1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit

Enter your choice: 4

Enter grade (A-F): C

Name: Wajiha Nazir, Roll: 3021, Marks: [99, 55, 68], Percentage: 74.00, Grade: C
Name: Rumaisa Nazir, Roll: 3082, Marks: [76, 85, 64], Percentage: 75.00, Grade: C

--- Student Record System ---

1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit

Student Name: _____ Wajiha Nazir _____ Roll No: _____ 25K-3021 _____

Section: BSE-1B

```
--- Student Record System ---
1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit
Enter your choice: 5
Class Average: 55.17
Students above average:
Name: Wajiha Nazir, Roll: 3021, Marks: [99, 55, 68], Percentage: 74.00, Grade: C
Name: Rumaisa Nazir, Roll: 3082, Marks: [76, 85, 64], Percentage: 75.00, Grade: C
Students below average:
Name: Tayyaba Nazir, Roll: 3096, Marks: [55, 23, 41], Percentage: 39.67, Grade: F
Name: Zunnera Nazir, Roll: 3013, Marks: [23, 32, 41], Percentage: 32.00, Grade: F

--- Student Record System ---
1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit
Enter your choice: 6
Ranked Students:
Rank 1: Name: Rumaisa Nazir, Roll: 3082, Marks: [76, 85, 64], Percentage: 75.00, Grade: C
Rank 2: Name: Wajiha Nazir, Roll: 3021, Marks: [99, 55, 68], Percentage: 74.00, Grade: C
Rank 3: Name: Tayyaba Nazir, Roll: 3096, Marks: [55, 23, 41], Percentage: 39.67, Grade: F
Rank 4: Name: Zunnera Nazir, Roll: 3013, Marks: [23, 32, 41], Percentage: 32.00, Grade: F

--- Student Record System ---
1. Add Student
2. Display All Students
3. Search Student by Roll or Name
4. Find Students by Grade
5. Class Average & Above/Below Average
6. Display Students by Rank
0. Exit
Enter your choice: 0
Exiting...

-----
Process exited after 546.4 seconds with return value 0
Press any key to continue . . .
```

Question 2:

Build an Bank Account Management system that stores.

- Account types: Savings, Current, Fixed Deposit
- Minimum balance requirements: Savings(\$1000), Current(\$5000), Fixed Deposit(\$10000)
- Interest rates: Savings(4%), Fixed Deposit(7%), Current(0%)
- Transaction history with date and type (Deposit/Withdrawal)
- Account creation date and last transaction date

Student Name: _____ Wajiha Nazir _____ Roll No: _____ 25K-3021 _____

Section: BSE-1B

```
#include <stdio.h>
#include <string.h>
#include <time.h>

#define MAX_ACCOUNTS 100

typedef struct {
    char date[11];
    char type[10];
    float amount;
} Transaction;

typedef struct {
    int accountNo;
    char name[50];
    char type[20];
    float balance;
    float interestRate;
    char creationDate[11];
    char lastTransactionDate[11];
    Transaction transactions[50];
    int transactionCount;
} Account;

Account accounts[MAX_ACCOUNTS];
int accountCount = 0;

void addAccount();
void deposit(Account *acc);
void withdraw(Account *acc);
void displayAccount(Account acc);
void showAllAccounts();
void updateDate(char *date);

int main() {
    int choice, i;
    do {
        printf("\n--- Bank Account System ---\n");
        printf("1. Add Account\n2. Deposit\n3. Withdraw\n4. Display All Accounts\n0. Exit\n");
        printf("Enter choice: "); scanf("%d",&choice);

        switch(choice){
            case 1: addAccount(); break;
            case 2: { int accNo; printf("Enter account no: "); scanf("%d",&accNo);
```

Student Name: Wajiha Nazir Roll No: 25K-3021

Section: BSE-1B

```

        for(i=0;i<accountCount;i++) if(accounts[i].accountNo==accNo) {withdraw(&accounts[i]); break;} } break;
        case 4: showAllAccounts(); break;
        case 0: printf("Exiting...\n"); break;
        default: printf("Invalid choice!\n");
    }
}while(choice!=0);
return 0;
}

void updateDate(char *date){
    time_t t = time(NULL);
    struct tm tm = *localtime(&t);
    sprintf(date,"%02d/%02d/%04d",tm.tm_mday, tm.tm_mon+1, tm.tm_year+1900);
}

void addAccount(){
    if(accountCount>=MAX_ACCOUNTS){ printf("Max accounts reached!\n"); return;}
    Account a;
    a.accountNo = accountCount+1;
    printf("Enter account holder name: "); getchar(); fgets(a.name,50,stdin); a.name[strcspn(a.name,"\n")]=0;
    int typeChoice;
    printf("Account Type: 1.Savings 2.Current 3.Fixed: "); scanf("%d",&typeChoice);
    if(typeChoice==1){ strcpy(a.type,"Savings"); a.balance=1000; a.interestRate=4; }
    else if(typeChoice==2){ strcpy(a.type,"Current"); a.balance=5000; a.interestRate=0; }
    else { strcpy(a.type,"Fixed Deposit"); a.balance=10000; a.interestRate=7; }
    updateDate(a.creationDate);
    strcpy(a.lastTransactionDate,a.creationDate);
    a.transactionCount=0;
    accounts[accountCount++]=a;
    printf("Account created successfully! Account no: %d\n", a.accountNo);
}

void deposit(Account *acc){
    float amount; printf("Enter deposit amount: "); scanf("%f",&amount);
    acc->balance+=amount;
    updateDate(acc->lastTransactionDate);
    Transaction t; strcpy(t.type,"Deposit"); t.amount=amount; updateDate(t.date);
    acc->transactions[acc->transactionCount++]=t;
    printf("Deposit successful! Current balance: %.2f\n", acc->balance);
}

void withdraw(Account *acc){
    float amount; printf("Enter withdrawal amount: "); scanf("%f",&amount);
    acc->balance+=amount;
    updateDate(acc->lastTransactionDate);
    Transaction t; strcpy(t.type,"Deposit"); t.amount=amount; updateDate(t.date);
    acc->transactions[acc->transactionCount++]=t;
    printf("Deposit successful! Current balance: %.2f\n", acc->balance);
}

void withdraw(Account *acc){
    float amount; printf("Enter withdrawal amount: "); scanf("%f",&amount);
    float minBalance = (strcmp(acc->type,"Savings")==0)?1000: (strcmp(acc->type,"Current")==0?5000:10000);
    if(acc->balance-amount<minBalance) {printf("Cannot withdraw. Minimum balance required!\n"); return;}
    acc->balance-=amount;
    updateDate(acc->lastTransactionDate);
    Transaction t; strcpy(t.type,"Withdrawal"); t.amount=amount; updateDate(t.date);
    acc->transactions[acc->transactionCount++]=t;
    printf("Withdrawal successful! Current balance: %.2f\n", acc->balance);
}

void displayAccount(Account acc){
    int i;
    printf("Account No: %d, Name: %s, Type: %s, Balance: %.2f, Interest: %.2f%%, Created: %s, Last Transaction: %s\n",
        acc.accountNo, acc.name, acc.type, acc.balance, acc.interestRate, acc.creationDate, acc.lastTransactionDate);
    printf("Transactions:\n");
    for(i=0;i<acc.transactionCount;i++)
        printf("%s - %s: %.2f\n", acc.transactions[i].date, acc.transactions[i].type, acc.transactions[i].amount);
}

void showAllAccounts(){
    int i;
    if(accountCount==0){ printf("No accounts to display.\n"); return;}
    for(i=0;i<accountCount;i++) displayAccount(accounts[i]);
}

```

Student Name: Wajiha Nazir Roll No: 25K-3021 Section: BSE-1B

```
--- Bank Account System ---
1. Add Account
2. Deposit
3. Withdraw
4. Display All Accounts
0. Exit
Enter choice: 1
Enter account holder name: Wajiha Nazir
Account Type: 1.Savings 2.Current 3.Fixed: 1
Account created successfully! Account no: 1

--- Bank Account System ---
1. Add Account
2. Deposit
3. Withdraw
4. Display All Accounts
0. Exit
Enter choice: 2
Enter account no: 1
Enter deposit amount: 500
Deposit successful! Current balance: 1500.00

--- Bank Account System ---
1. Add Account
2. Deposit
3. Withdraw
4. Display All Accounts
0. Exit
Enter choice: 3
Enter account no: 1
Enter withdrawal amount: 200
Withdrawal successful! Current balance: 1300.00

--- Bank Account System ---
1. Add Account
2. Deposit
3. Withdraw
4. Display All Accounts
0. Exit
Enter choice: 4
Account No: 1, Name: Wajiha Nazir, Type: Savings, Balance: 1300.00, Interest: 4.00%, Created: 29/11/2025, Last Transaction: 29/11/2025
Transactions:
29/11/2025 - Deposit: 500.00
29/11/2025 - Withdrawal: 200.00

--- Bank Account System ---
1. Add Account
2. Deposit
3. Withdraw
4. Display All Accounts
0. Exit
Enter choice: 0
Exiting...

-----
Process exited after 61.29 seconds with return value 0
Press any key to continue . . .
```

Question 3:

Build an employee record system that stores.

- Employee ID, name, department, designation, salary, date of joining, phone number, email
- Department categories: IT, HR, Finance, Marketing, Operations
- Designation hierarchy: Intern, Junior, Senior, Manager, Director
- Years of experience calculation based on joining date

Student Name: _____Wajiha Nazir_____ Roll No: _____25K-3021_____

Section: BSE-1B

Additional features:

- Department-wise salary statistics
- Annual appraisal with salary increment (5-15% based on performance)
- Search employees by department, designation, or experience
- Calculate and display employees eligible for promotion (>3 years in current role)

Student Name: Wajiha Nazir

Roll No: 25K-3021

Section: BSE-1B

```
#include <stdio.h>
#include <string.h>
#include <time.h>

#define MAX_EMPLOYEES 100

typedef struct {
    int id;
    char name[50];
    char department[20];
    char designation[20];
    float salary;
    int joinDay, joinMonth, joinYear;
    char phone[15];
    char email[50];
} Employee;

Employee employees[MAX_EMPLOYEES];
int employeeCount=0;

void addEmployee();
void displayEmployee(Employee e);
void departmentStats();
int experience(Employee e);
void promotionEligibility();
void searchEmployees();

int main(){
    int choice, i;
    do{
        printf("\n--- Employee Record System ---\n");
        printf("1.Add Employee\n2.Display All\n3.Department Stats\n4.Promotion Eligibility\n5.Search\n0.Exit\n");
        printf("Enter choice: "); scanf("%d",&choice);
        switch(choice){
            case 1: addEmployee(); break;
            case 2: for(i=0;i<employeeCount;i++) displayEmployee(employees[i]); break;
            case 3: departmentStats(); break;
            case 4: promotionEligibility(); break;
            case 5: searchEmployees(); break;
            case 0: printf("Exiting...\n"); break;
            default: printf("Invalid choice!\n"); break;
        }
    }while(choice!=0);
    return 0;
```

Student Name: _____ Wajiha Nazir _____ Roll No: _____ 25K-3021 _____

Section: BSE-1B

```

    }

void addEmployee() {
    if(employeeCount>=MAX_EMPLOYEES){ printf("Max employees reached!\n"); return;}
    Employee e;
    e.id = employeeCount+1;
    printf("Enter name: "); getchar(); fgets(e.name,50,stdin); e.name[strcspn(e.name,"\n")]=0;
    printf("Department (IT/HR/Finance/Marketing/Operations): "); scanf("%s",e.department);
    printf("Designation (Intern/Junior/Senior/Manager/Director): "); scanf("%s",e.designation);
    printf("Salary: "); scanf("%f",&e.salary);
    printf("Joining Date (DD MM YYYY): "); scanf("%d %d %d",&e.joinDay,&e.joinMonth,&e.joinYear);
    printf("Phone: "); scanf("%s",e.phone);
    printf("Email: "); scanf("%s",e.email);
    employees[employeeCount++]=e;
    printf("Employee added successfully!\n");
}

void displayEmployee(Employee e){
    printf("ID:%d, Name:%s, Dept:%s, Designation:%s, Salary:%.2f, Joined:%02d/%02d/%04d, Phone:%s, Email:%s, Exp:%d yrs\n",
        e.id,e.name,e.department,e.designation,e.salary,e.joinDay,e.joinMonth,e.joinYear,e.phone,e.email,experience(e));
}

int experience(Employee e){
    time_t t=time(NULL); struct tm tm=*localtime(&t);
    int years = tm.tm_year+1900 - e.joinYear;
    if(tm.tm_mon+1<e.joinMonth || (tm.tm_mon+1==e.joinMonth && tm.tm_mday<e.joinDay)) years--;
    return years;
}

void departmentStats(){
    int i;
    if(employeeCount==0){ printf("No employees.\n"); return;}
    char dept[20];
    printf("Enter department: "); scanf("%s",dept);
    int count=0; float total=0;
    for(i=0;i<employeeCount;i++){
        if(strcmp(employees[i].department,dept)==0){ total+=employees[i].salary; count++; }
    }
    if(count==0){printf("No employees in %s.\n",dept); return;}
    printf("Department: %s, Employees:%d, Avg Salary: %.2f\n",dept,count,total/count);
}

void promotionEligibility(){
    int i;
    if(employeeCount==0){ printf("No employees.\n"); return;}
    printf("Employees eligible for promotion (>3 years):\n");
    for(i=0;i<employeeCount;i++){
        if(experience(employees[i])>3) displayEmployee(employees[i]);
    }
}

void searchEmployees(){
    int i;
    if(employeeCount==0){ printf("No employees.\n"); return;}
    int choice;
    printf("Search by 1.Department 2.Designation 3.Experience: "); scanf("%d",&choice);
    if(choice==1){ char dept[20]; printf("Enter department: "); scanf("%s",dept);
        for(i=0;i<employeeCount;i++) if(strcmp(employees[i].department,dept)==0) displayEmployee(employees[i]); }
    else if(choice==2){ char des[20]; printf("Enter designation: "); scanf("%s",des);
        for(i=0;i<employeeCount;i++) if(strcmp(employees[i].designation,des)==0) displayEmployee(employees[i]); }
    else if(choice==3){ int yrs; printf("Enter experience in years: "); scanf("%d",&yrs);
        for(i=0;i<employeeCount;i++) if(experience(employees[i])>=yrs) displayEmployee(employees[i]); }
    else printf("Invalid choice!\n");
}

```

Student Name: Wajiha Nazir Roll No: 25K-3021 Section: BSE-1B

```
--- Employee Record System ---
1.Add Employee
2.Display All
3.Department Stats
4.Promotion Eligibility
5.Search
0.Exit
Enter choice: 1
Enter name: Wajiha Nazir
Department (IT/HR/Finance/Marketing/Operations): IT
Designation (Intern/Junior/Senior/Manager/Director): Director
Salary: 1000000
Joining Date (DD MM YYYY): 01 06 2021
Phone: 1234567890
Email: wajiha@gmail.com
Employee added successfully!

--- Employee Record System ---
1.Add Employee
2.Display All
3.Department Stats
4.Promotion Eligibility
5.Search
0.Exit
Enter choice: 2
ID:1, Name:Wajiha Nazir, Dept:IT, Designation:Director, Salary:1000000.00, Joined:01/06/2021, Phone:1234567890, Email:wajiha@gmail.com, Exp:4 yrs

--- Employee Record System ---
1.Add Employee
2.Display All
3.Department Stats
4.Promotion Eligibility
5.Search
0.Exit
Enter choice: 3
Enter department: IT
Department: IT, Employees:1, Avg Salary: 1000000.00

--- Employee Record System ---
1.Add Employee
2.Display All
3.Department Stats
4.Promotion Eligibility
5.Search
0.Exit
Enter choice: 4
Employees eligible for promotion (>3 years):
ID:1, Name:Wajiha Nazir, Dept:IT, Designation:Director, Salary:1000000.00, Joined:01/06/2021, Phone:1234567890, Email:wajiha@gmail.com, Exp:4 yrs

--- Employee Record System ---
1.Add Employee
2.Display All
3.Department Stats
4.Promotion Eligibility
5.Search
0.Exit
Enter choice: 5
Search by 1.Department 2.Designation 3.Experience: 2
Enter designation: Director
ID:1, Name:Wajiha Nazir, Dept:IT, Designation:Director, Salary:1000000.00, Joined:01/06/2021, Phone:1234567890, Email:wajiha@gmail.com, Exp:4 yrs

--- Employee Record System ---
1.Add Employee
2.Display All
3.Department Stats
4.Promotion Eligibility
5.Search
0.Exit
Enter choice: 0
Exiting...

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Process exited after 177.3 seconds with return value 0
Press any key to continue . . .
```